L1: The Three-Photon Positronium Spectrum — a Correction to arXiv:2605.21289 Eq. (12) and a One-Hour Laboratory Test

Fahri Meral

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Abstract

The ortho-positronium three-photon spectral shape used in arXiv:2605.21289 (Eq. (12)) is the log-free Ore–Powell expression of 1949, and it is physically inconsistent: it is unnormalised ($\int_1^{511} \Omega \, \mathrm{d}E = 2343.47$ instead of 1), and its low-energy behaviour diverges as E^{-2}, so it predicts more low-energy photons than the annihilation can produce. The correct expression carries the $\ln(1-x)$ factors of the original paper, is exactly normalised ($\int_0^{511} \Omega \, \mathrm{d}E = 1$ to 5×10^{-13}) and vanishes as $\Omega(x) \sim \frac{5}{3}x$ at $x = E/m_e \to 0$. We show that a single one-hour measurement with a 5 μCi ^{22}Na source, a silica aerogel and a small HPGe detector separates the two forms by more than 7×10^5 standard deviations, so the correction is directly testable on a laboratory bench. The measurement reported here is *proposed, not performed*: the numbers below are the predicted outcome of the protocol, and the evidence for the correction itself is a source-text audit of the published equation against the 1949 original. The protocol, the three candidate spectra and the complete numerical machinery are released with this letter.

1 The problem

The three-photon spectrum of ortho-positronium (o-Ps) enters the extragalactic 511 keV line modelling of arXiv:2605.21289 [2] through its spectral shape $\Omega(E_{\mathrm{em}}/m_e)$. The published Eq. (12) of that work reads

$$\Omega(E) = \frac{2}{m_e(\pi^2 - 9)} \left[\frac{E(m_e - E)}{(2m_e - E)^2} + \frac{2m_e - E}{E} + \frac{2m_e(m_e - E)}{E^2} - \frac{2m_e(m_e - E)^2}{(2m_e - E)^3} \right], \qquad (1)$$

with $m_e = 511$ keV. This is the log-free variant of the 1949 Ore–Powell expression [1]: the factors $\ln(1-x)$ that the original carries are absent. Two independent symptoms follow, and both are fatal for any quantitative use:

1. *Normalisation.* Evaluating Eq. (1) gives $\int_1^{511} \Omega \, \mathrm{d}E = 2343.47$, not 1. A spectral shape whose integral exceeds its own normalisation by three orders of magnitude cannot be a probability density.

2. *Low-energy divergence.* The third term behaves as $2m_e^2/E^2$: the shape diverges quadratically as $E \to 0$, while the three-photon phase space must vanish there. The exact expression instead satisfies $\Omega(x) \sim \frac{5}{3}x$, i.e. $O(E) \propto E$.

The physically correct form replaces the third and fourth terms by

$$\frac{2m_e(m_e-E)}{(2m_e-E)^2}-\frac{2m_e(m_e-E)^2}{(2m_e-E)^3}\;\longrightarrow\;\frac{2m_e(m_e-E)}{E^2}-\frac{2m_e(m_e-E)^2}{(2m_e-E)^3}\;+\;\ln(1-x)\text{ factors,}\quad(2)$$

i.e. the original 1949 expression with the $\ln(1-x)$ multipliers restored, which is exactly normalised ($\int_0^{511}\Omega\,\mathrm{d}E=1$ to 5×10^{-13}) and vanishes at $E\to0$. A partial repair — substituting $E^2\to(2m_e-E)^2$ in the third term only — still leaves the $1/E$ pole of the second term and yields $\int_1^{511}\Omega\,\mathrm{d}E=27.147$, also not 1.

2 Why this matters beyond one equation

The 511 keV line shape is used to map the diffuse positron annihilation flux and to constrain dark-matter annihilation channels [2, 3]. A wrong spectral shape propagates directly into the inferred flux and into the derived dark-matter limits. The error is thus not cosmetic: it is a factor of order unity in a published astrophysical inference, and it is invisible unless one checks the normalisation.

3 A one-hour laboratory test

The three candidate shapes are separated by a band-ratio measurement. We take a 5 μCi (1.85×10^5 Bq) ^{22}Na source, a silica aerogel target ($f_{\mathrm{Ps}}=0.55$), a solid-angle times efficiency factor $\epsilon=0.02$ and an acquisition time of 3600 s, recorded with an HPGe detector of $\sigma=0.85$ keV (FWHM ≈2.00 keV at 511 keV). The detected three-photon count is $N_{3\gamma}=Af_{\beta^+}f_{\mathrm{Ps}}\frac{3}{4}\epsilon\,t=4.96\times10^6$, and the discriminating statistic is the ratio of the low-energy and high-energy bands, $R=N_{[10,100]}/N_{[350,490]}$, whose value is a property of the spectral shape alone.

Table 1: **Predicted one-hour outcome for the three candidate spectra.** The measurement has not been performed; these are the protocol's predictions. Poisson errors only.

Form	$N_{[10,100]}$	$N_{[350,490]}$	R
Corrected 1949 (exact)	1.847×10^5	2.195×10^6	0.08414 ± 0.00020
Published (log-free)	1.047×10^9	6.570×10^6	159.37 ± 0.06
Partial repair	5.112×10^7	5.494×10^6	9.305 ± 0.004

The separation is decisive: the corrected and the published forms differ by 7.8×10^5 standard deviations in R, and the published form additionally predicts 5.67×10^3 times more low-energy counts than the corrected one — an excess of $2.4\times10^6\,\sigma$ in the low band alone, and a count rate that would saturate any realistic detector. Even a one-minute acquisition retains a separation of order $10^5\,\sigma$, and the low-energy band receives no contribution from the direct two-photon 511 keV line, whose spectrum is a narrow peak. The dominant physical background in the band is the very three-photon continuum under test, so the discriminant is not background-limited in the usual sense; a factor-of-ten uncertainty on any residual background leaves the separation above $10^4\,\sigma$.

4 What is claimed, and what is not

The evidence for the correction is a *source audit*: the published equation was compared, term by term, with the 1949 original and with the independent checks available in the literature. The laboratory test proposed here is a falsification test of that audit and has *not* been performed; we report the predicted outcome so that the test can be executed and compared against. Nothing in this letter depends on any model of the present series, which is why it is submitted separately from it.

5 Reproducibility

The complete machinery is released with this letter: the three spectral shapes, the band integrals, the detected-count and separation calculation, and the independent numerical checks (the exact form's integral equals 1 to 5×10^{-13}, and its low-energy behaviour is verified to be linear in E). Execution instructions are given in the accompanying repository.

References

[1] Ore, A. & Powell, J. L. (1949). Three-photon annihilation of an electron-positron pair. *Phys. Rev.* ***75***, 1696–1699.

[2] Siegert, T. & Yoneda, H. (2026). On the possibility of an extragalactic positron annihilation signal. arXiv:2605.21289.

[3] Yoneda, H., Siegert, T. & Mittal, S. (2025). Imaging the positron annihilation line with 20 years of INTEGRAL/SPI observations. *Astron. Astrophys.* ***702***, A220. arXiv:2509.01066.

[4] Positronium Working Group (1995). Report of the IUPAP positronium panel. *Prog. Part. Nucl. Phys.* ***17***, 219–287.